

# Wonky Kernel

A code-first CAD kernel for programmable modeling, automated checks and LLM-assisted workflows.

|                                    |                                                     |                                                        |
|------------------------------------|-----------------------------------------------------|--------------------------------------------------------|
| <b>Bend</b><br>Geometry & topology | <b>FeatureScript + Python</b><br>Modeling frontends | <b>Analytic B-rep</b><br>Structured, exportable solids |
|------------------------------------|-----------------------------------------------------|--------------------------------------------------------|

AVAILABLE TODAY

## Modeling input

Original Onshape **FeatureScript syntax** with a growing library subset. Real Python execution with a limited **build123d algebra API**.

## Boolean coverage

Coaxial-cylinder union, intersection and difference; bounded planar union/difference; admitted plane/cylinder intersections with convex planar tools.

## Export & validation

Analytic **STEP** and structured **B-rep JSON**. Native tests, independent STEP checks, geometry comparisons and standard-view image diffs.

## Geometry & operations

Planar profiles, line/arc extrusions, cylinders, conical frustums, rigid transforms and frozen analytic Onshape body imports.

## Implementation & numerics

Geometry in **Bend**; JavaScript handles interpretation and I/O. F32x2 arithmetic, selected exact predicates and explicit error budgets. No OCCT production fallback.

## Inspection & feedback

Browser viewer with selection, hover, side-by-side and slider comparison, plus saved annotations. Geometry-to-code provenance, named sketch roles and operation ancestry.

VERIFIED MILESTONE

**12 original arcs → one closed analytic solid**

The original M3 profile runs through FeatureScript and exports without faceting. Real r10b unions **g7** and **g9** pass; the full model currently stops at the subsequent curved-body union (**g10**).

## Current boundaries

General curved Booleans, fillets, revolve, enclosed cavity shells and full API coverage remain open. Unsupported geometry fails explicitly.

## Performance & next work

The current Bend/JS modeling path is slower than build123d/OCCT in six verified Python workloads. Broader Booleans and performance improvements are in progress.

Scope: implemented, bounded operation families. Active development; not a production-complete general CAD kernel.